

Capacity Achieving Cyclic Codes

Daniel Welchons¹

¹Benedictine College*

April 24, 2026

Abstract

While the existence of families of capacity achieving codes has been known since Shannon's seminal paper, finding capacity achieving codes is a notoriously hard problem. Since 2016 it has been known that the families of codes with permutation groups that are "close" to doubly transitive achieve capacity on the q -ary erasure channel under symbol MAP decoding. In particular, cyclic codes which naturally have transitive permutation groups can be shown to be capacity achieving based on the size of the first cyclotomic coset.

This paper focuses on reducing the needed condition to one on the length of the codes and shows that such codes exist for any rational rate. Moreover, we provide a constructive method for creating a family of capacity achieving codes for any rational rate.

Key Words : Cyclic Codes, Capacity Achieving, Permutation Groups, Coding theory, Cryptography.

1 Introduction

When Shannon defined channel capacity as the maximum information per symbol that can reliably be sent over a channel, he also showed that, for any capacity, there exist capacity achieving codes [Sha48]. However, he used a probabilistic argument to show such codes exist, meaning he gave no explicit examples. Finding explicit examples of capacity achieving codes is a notoriously difficult problem. Indeed, the first family of linear codes shown to be capacity achieving appeared in [RSU01], more than 50 years after Shannon proved the existence of such codes.

In 2016, the authors of [KKM⁺17] showed that on erasure channels families of liner codes (hereafter, just codes) with doubly transitive permutation

*dwelchons@benedictine.edu

groups are capacity achieving codes—settling a long standing conjecture about Reed-Muller codes in the affirmative. In the same year, this was improved by [KCP16], by weakening the double transitivity condition. Both papers rely on the properties of the average conditional entropy of each position (through the average Extrinsic Information Transfer function, average EXIT). They use the fact that for any code of rate r the average EXIT function transitions from 0 to 1 when the erasure rate reaches $1 - r$. Notably, for a family of codes with rates converging to $r \in (0, 1)$, then, if the space over which this transition occurs converges to 0 (i.e. the transition is sufficiently sharp), then the family will be capacity achieving [KKM⁺17].

Specifically, in [KKM⁺17], they showed that the strong symmetry properties imposed by double transitivity ensure that such a sharp transition occurs. Hence, families of codes with doubly transitive permutation groups are capacity achieving on the binary erasure channel (BEC) under symbol-wise *Maximum a Posteriori* (MAP) decoding (while inefficient, MAP decoding is very robust and represents something of a best case for code performance). This finding was notably applied to Reed-Muller codes and a number of other classic families of codes which have doubly transitive permutation groups or are very closely related to families with doubly transitive permutation groups [KKM⁺17].

The authors of [KCP16] showed that double transitivity of the permutation group is too strong of a symmetry requirement and that the sharp transition occurs under more relaxed conditions. Instead of the double transitivity requirement, transitivity and increasing minimum orbit lengths under the action of the stabilizer of the first position is sufficient. Although cumbersome to state, the conditions of [KCP16], amount to a statement that the family of codes needs a stronger symmetry than simple transitivity of the permutation groups, but need not have double transitivity. This is because doubly transitive permutation groups correspond to largest possible minimum orbit lengths under the action of the stabilizer of the first position. Additionally, they were able to remove the binary restriction.

Since cyclic codes have permutation groups which are always transitive, they are a natural place to apply these results. The techniques that [KKM⁺17] and [KCP16] developed have been applied to certain Abelian codes in [NK23], which are a more general class of codes. However the results of [NK23] apply only to families of codes with fixed sets of prime divisors.

This paper seeks to extend the results of [KCP16] by identifying simplified conditions for when cyclic codes achieve capacity. Although we use the same machinery as [KCP16], we develop a condition on when these techniques apply to cyclic codes which is more broad than that of [KCP16] and disjoint from those of [NK23]. Additionally, we establish a technique to construct families of capacity achieving cyclic codes of any rate.

2 Preliminaries

We will adopt the definition of capacity achieving codes on q -ary erasure channel used in [KKM⁺17] and [KCP16]. Namely, a family of q -ary codes $\{\mathcal{C}_n\}$ with rates converging to r is capacity achieving on the q -ary erasure channel if for every erasure probability $p < 1 - r$ the probability of decoding failure converges to zero as $n \rightarrow \infty$. This formulation is equivalent to [Sha48], but is more succinct for our application. Further, we adopt the notation $[n] = \{1, 2, \dots, n\}$ for any $n \in \mathbb{N}$.

As the name suggests cyclic codes are characterized by the fact that for any codeword $c = [c_0, c_1, \dots, c_{n-2}, c_{n-1}]$ the vector $[c_{n-1}, c_0, c_1, \dots, c_{n-2}]$ is also a codeword of $\mathcal{C}$. While this is a useful and intuitive definition, we can bring a much larger body of algebraic work to bear by noting that a q -ary $[n, k]$ cyclic code can be interpreted as an ideal of $\mathbb{F}_q[x]/(x^n - 1)$ generated by some factor g of $x^n - 1$ such that g has degree $n - k$.

The ideals of $\mathbb{F}_q[x]/(x^n - 1)$ correspond to factors of $x^n - 1$ over $\mathbb{F}_q$. Moreover, $x^n - 1$ has no repeated irreducible factors if and only $\gcd(n, q) = 1$. So in the case that n is relatively prime to q , there is a bijection between cyclic codes and factors of $x^n - 1$. Further, we can say more about the permutation groups of the codes in the case that n and q are relatively prime. Hence, we restrict ourselves to lengths which are relatively prime to the size of the field. In particular, we consider binary codes of odd length, though some of the theorems apply as long as the length of the code is relatively prime to the order of the field.

Since the permutation group is the primary tool of analysis for this paper, we will now turn our attention to defining the permutation group of a code along with its key properties.

Given a code $\mathcal{C}$ of length n , the *permutation group* of $\mathcal{C}$ is defined to be:

$$\text{Perm}(\mathcal{C}) = \{P \in S_n \mid \forall c \in \mathcal{C}, cP \in \mathcal{C}\},$$

where S_n is the symmetric group on n elements.

We may interpret $\text{Perm}(\mathcal{C})$ as either a group of $n \times n$ permutation matrices or as permutations shuffling the entries of the vectors. Both are equivalent, but we will tend toward the latter in this paper. Even though the permutation group is defined based on codewords being mapped to other codeword, the action of the group on the coordinates of the codewords of $\mathcal{C}$ is more important. Hence, we define orbits and transitivity based on the coordinates of the codewords in $\mathcal{C}$.

For any position i with $0 \leq i \leq n - 1$ the *orbit* of i is the set of positions j for which there exists a permutation in $\text{Perm}(\mathcal{C})$ which maps i to j . Moreover, the *length* or *size* of the orbit of i is the cardinality of the orbit of i as a set. For any position i with $0 \leq i \leq n - 1$ the *stabilizer* of i is the set of permutations in $\text{Perm}(\mathcal{C})$ which map position i to itself, denoted as $\text{Stab}_i(\mathcal{C})$.

A permutation group, $\text{Perm}(\mathcal{C})$, is said to be *transitive* if for every pair i, j such that $0 \leq i, j \leq n-1$ there is some $P \in \text{Perm}(\mathcal{C})$ such that position i is mapped to position j under the action of P . Further if there is some i such that $\text{Stab}_i(\mathcal{C})$ is also transitive on the remaining $n-1$ positions, then $\text{Perm}(\mathcal{C})$ is *doubly transitive*. In the case of a transitive permutation group, all of the stabilizers are conjugate groups, so it is typical to say that $\text{Perm}(\mathcal{C})$ is doubly transitive if $\text{Stab}_0(\mathcal{C})$ is transitive on $[n-1]$.

In the case of cyclic codes, there are two permutations which are particularly important. The first is the cyclic shift and the other arises from the Frobenius automorphism. Namely, in a field of characteristic p , $(a+b)^p = a^p + b^p$. Using this fact, it follows that in $\mathbb{F}_q[x]$, $f(x^q) = f(x)^q$. Importantly, in $\mathbb{F}_q[x]/(x^n - 1)$ with n relatively prime to q , $f(x^q)$ corresponds to the permutation $i \mapsto iq \pmod n$ (this is a permutation when n is relatively prime to q). Hence, because $f(x^q) = f(x)^q$, we see that if f is in an ideal I , then $f(x^q)$ is also in I , meaning that $i \mapsto iq \pmod n$ is in the permutation group of any cyclic code of length n . We will refer to the cyclic shift and the Frobenius permutation by $\pi : i \mapsto i+1 \pmod n$ and $\phi : i \mapsto iq \pmod n$, respectively.

We may now state the main tool that we will use to prove the results of this paper. Let m_n be the minimum orbit length induced by $\text{Stab}_0(\mathcal{C}_n)$ for any $i \in [n-1]$.

Theorem 1 ([KCP16]). *Let $\{\mathcal{C}_n\}$ be a sequence of codes with increasing blocklengths, where the rates satisfy $r_n \rightarrow r \in (0, 1)$. If $\text{Perm}(\mathcal{C}_n)$ is transitive for each n and $m_n \rightarrow \infty$ then $\{\mathcal{C}_n\}$ is capacity achieving on the q -ary erasure channel under symbol-MAP decoding.*

We will refer to the transitive permutation group and increasing m_n of the proceeding theorem as the Transitivity + Orbit condition. Extending the general result to cyclic codes specifically:

Theorem 2 ([KCP16]). *Let $\{\mathcal{C}_n\}$ be a sequence of $\mathbb{F}_q$ cyclic codes with rates $r_n \rightarrow r \in (0, 1)$ and increasing blocklengths N_n . Suppose N_n divides $q^{t_n} - 1$ (where t_n is the least integer for which this holds) and is relatively prime to $q-1$, and let s_n be the smallest prime divisor of t_n . If $s_n \rightarrow \infty$, then $\{\mathcal{C}_n\}$ is capacity achieving on the q -ary erasure channel under MAP-decoding.*

Theorem 1 will provide us with our main tool for establishing families of codes as capacity achieving, but we will extend the cyclic code results beyond that of Theorem 2. In particular, understanding the orbits induced by the action of ϕ will allow us to extend these results.

The cyclic shift π guarantees that the permutation group of any cyclic code is transitive. On the other hand, ϕ fixes the first position. Thus, if the orbits that ϕ induces in the other positions are growing for some family of cyclic codes, then we will be able to apply Theorem 1. However, there is a

tradeoff in that ϕ must induce large enough minimum orbits, but as we shall see, it is important that the maximum orbit size not be too large relative to the blocklength in order to ensure codes of the desired rates exist.

3 Main Results

In order to establish the Transitivity + Orbit condition it is important to gain a lower bound on the orbits induced by ϕ . Moreover, the length the orbit of i under ϕ is the same as the size of the cyclotomic coset containing i . Indeed, this fact plays a key role in [KCP16] and the proof of Theorem 2. That is, the fact that the size of each cyclotomic coset must divide t_n gives a lower bound on the size of orbits which are induced by ϕ which in turn ensures that the Transitivity + Orbit condition is met [KCP16].

However, instead of examining the factorization of t_n (which depends on first finding the order of ϕ), this can be done by looking at the factorization of n . For this next section we will need to consider ϕ relative to multiple lengths. Hence, we will use a subscript to denote the modulus. That is $\phi_n : i \mapsto iq \pmod n$. Moreover, to avoid the clutter of saying “non-zero” every time that we discuss orbits, we will always consider ϕ_n as inducing orbits in the set $[n - 1]$.

Lemma 1. *Let $n = p_1^{a_1} \cdots p_k^{a_k}$ where p_1 through p_k are distinct primes and let $q = p^m$ such that $\gcd(n, q) = 1$. Then the smallest orbit induced by ϕ_n has the same length as the least orbit induced by ϕ_{p_i} for any p_i .*

Proof: Without loss of generality, suppose that p_1 induces the least orbit and let $i \in [p_1 - 1]$ such that i has minimum orbit length under ϕ_{p_1} and use b to denote that orbit length. Applying the definition of congruency mod p_1 , this implies that $q^b i = p_1 d + i$ for some integer d . Thus, $q^b i \frac{n}{p_1} = (p_1 d + i) \frac{n}{p_1} = nd + i \frac{n}{p_1}$, which means that the order of $i \frac{n}{p_1}$ is at most b under ϕ_n .

On the other hand, suppose that $i \in [n - 1]$ has least orbit length under ϕ_n , say b' . Define x and ℓ such that $x = \gcd(n, i)$ and $\ell = \frac{i}{x}$. By the structure of n it follows that $x = p_1^{t_1} \cdots p_k^{t_k}$ where $t_j \leq a_j$ for every $j \in [k]$. Moreover, there is some j such that $t_j < a_j$ as otherwise $i > n - 1$. Thus, $\frac{n}{x}$ is divisible by p_j , but does not divide ℓ .

Applying the definition of congruency mod n we see $q^{b'} i = nd + i$ which implies $q^{b'} \ell x = nd + \ell x$. Hence, $q^{b'} \ell = \frac{n}{x} d + \ell$ for some $d \in \mathbb{Z}$. Now by construction p_j divides x and not ℓ . Thus, $\ell \pmod{p_j} \neq 0$ and the order of $\ell \pmod{p_j}$ under ϕ_{p_j} is at most b' . However, we already established that the least orbit length under any of the prime divisors of n is b . Thus, $b' \geq b$. Therefore, by the minimality of b' and the fact we have elements of order at most b under ϕ_n , it follows that $b' = b$. $\square$

In addition to having the least orbit under the Frobenius map for composite n , we may provide a lower bound for the least orbit for a prime length.

Lemma 2. *Let p be a prime and q a power of a different prime. Suppose that t is the largest integer such that $q^t < p$. Then for any $a \in [p-1]$ the order of a is at least t under ϕ_p .*

Proof: Let $a \in [p-1]$ have least order, say b . Now a and q are relatively prime to p . Thus, $q^b a \bmod p = a$ implies that $q^b \bmod p = 1$. Note that the order of 1 is at least t as $1, q, \dots, q^t$ are all values less than p and therefore distinct elements of the orbit ϕ_p . Hence, $b \geq t$. $\square$

The actual orbit induced by a prime may be much larger (up to $p-1$, by Fermat's Little Theorem), but we know that if a prime is much larger than the base q , then we are guaranteed an orbit that is not too small. That is, for a sequence of increasing primes and fixed q the minimum orbit length under ϕ_n will increase to infinity, but may not always be strictly increasing. For instance, if $q = 2$, then $n = 11$ will result in elements of order 10, but if $n = 31 = 2^5 - 1$ then we will have elements of order 5. However, by Lemma 2, $n = 2^t - 1$ has minimum order of at least t , so overall the minimum order is increasing as $t \rightarrow \infty$.

We may combine these two lemmas to produce a simple characterization of families of cyclic codes which possess the Transitivity + Orbit condition based only on the least prime divisor of n . This result is very similar to that given in [KCP16], but it simplifies the conditions to require no exponentiation.

Theorem 3. *Let $\{\mathcal{C}_n\}$ be a family of q -ary cyclic codes with increasing block lengths N_n and rates $r_n \rightarrow r \in (0, 1)$. Let p_n be the least prime divisor of N_n . If $p_n \rightarrow \infty$ then $\{\mathcal{C}_n\}$ is capacity achieving on the q -ary erasure channel under MAP-decoding.*

Proof: Since the codes are all cyclic, transitivity is immediate. Hence, it suffices to show that ϕ_{N_n} induces increasing minimum orbits as it is a permutation which fixes the first coordinate. By Lemma 1, the length of the smallest orbit for any of the N_n positions is that of the smallest orbit induced on one of the prime divisors of N_n . For any prime p dividing N_n , the largest t for which $q^t < p$ is a lower bound on the orbit sizes induced by ϕ_p . Specifically, we define t_n to be the largest integer such that $q^{t_n} < p_n$. Since p_n is the least prime dividing N_n , t_n is the smallest lower bound obtained by applying Lemma 2 to each of the prime factors of N_n . Therefore, the smallest possible orbit under the action of ϕ_{N_n} has size t_n .

By assumption, $p_n \rightarrow \infty$ and q is fixed, meaning that $t_n \rightarrow \infty$. Therefore $\{\mathcal{C}_n\}$ satisfies the Transitivity + Orbit condition and is capacity achieving on the q -ary erasure channel under MAP-decoding. $\square$

Theorem 3 extends the bounds of Theorem 2 in two ways. Firstly, it reduces the condition from first finding a factor of n and then factoring the exponent to simply finding the least factor of n . Secondly, it can be shown that there are cases in which Theorem 3 applies, but Theorem 2 does not apply [Wel25].

We will now turn our attention to showing that there are capacity achieving codes which satisfy Theorem 3. The permutation ϕ allows us to establish a lower bound on the smallest orbits, but the order of ϕ determines the degree of the largest irreducible factor of $x^n - 1$. Hence, it determines what rates are achievable for any n . Thus, we hope to find a sequence of codes which has increasing minimum prime divisors, but the order of ϕ_n increases slowly with respect to n . In fact, the order of ϕ_n is the least integer m such that $n|q^m - 1$ which is also the size of the first and largest cyclotomic coset and the largest degree of an irreducible factor of $x^n - 1$ [HP03]. We will denote this order by $\text{ord}_n(q)$. Up until this point, the theory has applied in the general q -ary code case. However, the remaining work will focus on binary codes.

Lemma 3. *Let $\{N_n\}$ be a sequence of increasing odd integers with $m_n = \text{ord}_{N_n}(2)$. If $\left\{\frac{m_n}{N_n}\right\} \rightarrow 0$ then for any rational $r \in (0, 1)$ there exists a family of cyclic codes $\{\mathcal{C}_n\}$ with lengths $\{N_n\}$ and rates converging to r .*

Proof: Since r is a rational number it can be written as a fraction. However, since the rate of a cyclic code is $1 - \frac{\deg(g)}{N_n}$, where g is the generator polynomial of $\mathcal{C}$, it will be more useful to write $r = 1 - \frac{a}{b}$ with $a, b \in \mathbb{N}$. Define b sets as $P_1 = \dots = P_b = \{1\}$. We obtain b corresponding polynomials which multiply to $x^{N_n} - 1$ via the following process.

Let $M_1(x), \dots, M_{\ell_n}(x)$ be the irreducible factors of $x^{N_n} - 1$ with decreasing degree. For each $j \in [b]$, define $p_j = \prod_{M \in P_j} M$. We proceed iteratively for ℓ_n iterations such that in step i

1. Select the least j such that $\deg(p_j)$ is minimal over all $j \in [b]$
2. Add M_i as an element of P_j .
3. Update p_j

At the end of this process, we are left with b polynomials $p_1, \dots, p_b$ which have weakly decreasing degrees. Moreover, because at each step we assigned a polynomial of degree at most m_n to the set with the smallest polynomial degree we obtain factors such that $\deg(p_b) \leq \deg(p_1) \leq \deg(p_b) + m_n$.

We therefore construct $\mathcal{C}_n$ with generator polynomial $g(x) = p_1 \cdots p_a$. Since $g(x)$ consists of the a polynomials with largest degrees of the b factors, it follows that $\frac{\deg(g)}{N_n} > \frac{a}{b}$. Thus, $r_n = 1 - \frac{\deg(g)}{N_n} \leq 1 - \frac{a}{b} = r$. On the other hand, each of the a factors has degree no more than $\deg(p_b) + m_n$. Hence,

$$r_n = 1 - \frac{\deg(g)}{N_n} \geq 1 - \frac{a(\deg(p_b) + m_n)}{N_n} \geq 1 - \frac{a \deg(p_b)}{b \deg(p_b)} - \frac{am_n}{N_n} = r - \frac{am_n}{N_n}$$

From this we can conclude $r_n \rightarrow r$ as n increases since $\frac{m_n}{N_n} \rightarrow 0$. $\square$

Lemma 3 establishes the existence of codes with desirable rates, but does not show that such a sequence of codes will satisfy Theorem 2 or Theorem 3. We will now turn our attention to finding a sequence of integers which satisfies both Lemma 3 and Theorem 3, but we will first state a result which is a consequence of Lemma 3 and Theorem 2.

Corollary 1. *Let r be a rational rate in $(0, 1)$ and $\{m_n\}$ be an increasing sequence with increasing minimal prime divisors. Then there exists a sequence of cyclic codes $\{\mathcal{C}_n\}$ with length $2^{m_n} - 1$ with rates converging to r which is capacity achieving on the BEC under bit-MAP decoding.*

Proof: Since $N_n = 2^{m_n} - 1$ we have $\frac{m_n}{N_n} = \frac{m_n}{2^{m_n} - 1} \rightarrow 0$ as $n \rightarrow \infty$. Thus, by Lemma 3, there is a sequence of codes with lengths N_n and rates converging to r . By construction, the minimum prime divisor of m_n is increasing. Thus, Theorem 2 applies, and the sequence of codes must be capacity achieving on the BEC under bit-MAP decoding. $\square$

Let $\{s_n\}$ be an increasing sequence of integers. A prime p is a *primitive prime* divisor of s_m if $p \mid s_m$ and $p \nmid s_k$ for any $k < m$. Primitive primes are going to be the key to our approach because they will allow us to control $\text{ord}_n(2)$ while constructing a sequence of increasing integers. Importantly, the following theorem shows that such primitive primes exist for the sequence $2^n - 1$ as $n \rightarrow \infty$.

Theorem 4 (Zsigmondy Theorem [Zsi82, Wei24]). *If $1 \leq b < a$ such that $\gcd(a, b) = 1$ then each term in the sequence $\{a^n - b^n\}$ has at least one primitive prime factor except when:*

1. $a = 2, b = 1$ and $n = 6$
2. $n = 2$ and $a + b = 2^k$ for some $k \in \mathbb{N}$

Similarly $a^n + b^n$ has at least one primitive prime factor except in the case of $2^3 + 1^3$.

Zsigmondy's Theorem ensures that we have primes which will allow us to control the value of $\text{ord}_{N_n}(2)$. However, to ensure that the length is growing with respect to the order we need to note a few facts. Firstly, if p is primitive to $2^m - 1$ then $p > m$, which is a direct consequence of Fermat's Little Theorem. Secondly, if m_n is composite then it has a proper factor b_n which is at least $\sqrt{m_n}$. In turn, $2^{b_n} - 1$ must have primitive primes. Finally,

any primitive primes of $2^{b_n} - 1$ must be greater than b_n which gives us $p > b_n \geq \sqrt{m_n} > \lg(m_n)$ (with $\lg(x) = \log_2(x)$). With these facts in mind, we can now show the existence of a family of capacity achieving codes.

Theorem 5. *Let $\{m_n\}$ be an increasing sequence of composite natural numbers. Define $\{b_n\}$ to be a sequence of proper divisors of $\{m_n\}$ such that $b_n \geq \sqrt{m_n}$, and A_n to be the set of primes which are less than $\lg(m_n)$. Let $\{N_n\}$ be a sequence of integers such that N_n is a factor of $2^{m_n} - 1$ which is divisible by at least one of the primitive primes of $2^{m_n} - 1$, one primitive prime of $2^{b_n} - 1$, and is relatively prime to a for all $a \in A_n$.*

For any rational $r \in (0, 1)$ there exists a family of cyclic codes $\{\mathcal{C}_n\}$ with lengths N_n such that $\{\mathcal{C}_n\}$ has rates converging to r and is capacity achieving on the BEC under MAP decoding.

Proof: Note that the primitive primes of both $2^{m_n} - 1$ and $2^{b_n} - 1$ are greater than $\lg(m_n)$, which means that neither is in A_n . Moreover, because $b_n | m_n$ it follows that $2^{b_n} - 1 | 2^{m_n} - 1$ [MS77]. Thus, it is always possible to pick a length N_n that satisfies the conditions of the theorem.

The least primitive primes of $2^{m_n} - 1$ must increase to infinity as any p cannot be primitive for any $m_n > p - 1$. Hence, $\{N_n\} \rightarrow \infty$ as $n \rightarrow \infty$. Further, the construction removes “small” non-primitive primes, meaning that the minimum prime divisors of N_n are also increasing.

It now suffices to show that $\lim_{n \rightarrow \infty} \frac{m_n}{N_n} = 0$ in order to apply Lemma 3. Since, N_n divides $2^{m_n} - 1$, $\text{ord}_{N_n}(2) \leq m_n$. Because N_n is divisible by a primitive prime of $2^{m_n} - 1$ it follows that $\text{ord}_{N_n}(2) \geq m_n$. Therefore, $\text{ord}_{N_n}(2) = m_n$. Further, we know that $N_n > m_n \sqrt{m_n}$ as the known prime divisors are larger than m_n and $\sqrt{m_n}$ respectively. Thus,

$$\lim_{n \rightarrow \infty} \frac{m_n}{N_n} \leq \lim_{n \rightarrow \infty} \frac{m_n}{m_n \sqrt{m_n}} = 0.$$

Therefore, we may apply Lemma 3 to show that there exists a family of cyclic codes $\{\mathcal{C}_n\}$ with lengths $\{N_n\}$ and rates converging to r . Moreover, we have already established that Theorem 3 applies. Hence, $\{\mathcal{C}_n\}$ is capacity achieving on the BEC under bit-MAP decoding. $\square$

Because every odd n has some m for which $n | 2^m - 1$, Theorem 5 is not too restrictive. It is essentially saying that if we pick a sequence of increasing orders $\{m_n\}$, we may choose a sequence of integers $\{N_n\}$ such that $\text{ord}_{N_n}(2) = m_n$, which admit capacity achieving codes of any rational rate. Moreover, it tells us what factors we need for any given order. Further, together with Lemma 3, we know how to build codes of rates close to r when $N_n \gg m_n$. Thus, we not only know what lengths admit a family of capacity achieving codes, but we also know how to construct those codes.

4 Conclusions

Theorems 3 and 5 give us two ways to search for capacity achieving codes. Given a family of codes with non-trivial rates, Theorem 3 gives us a method to determine if the family is capacity achieving without needing to compute the order of each blocklength. On the other hand, given a sequence of increasing (non-prime) orders we can work backward, using Theorem 5, to find a sequence of capacity achieving codes regardless of the desired rate. Hence, we not only have a method evaluating families of codes, but we also have a method of constructing capacity achieving cyclic codes for a broad range of lengths and rates. Nevertheless, a complete characterization of capacity achieving cyclic codes remains open. Notably, for any sequence of lengths which do not satisfy Theorem 3, we can construct a sequence of non-capacity achieving codes with non-vanishing rates [Wel25], but [NK23] shows that there are many capacity achieving cyclic codes which do not satisfy Theorem 3. Hence, for families of codes with some repeated prime factor, there is a mixture of capacity and non-capacity achieving codes which is yet to be fully resolved.

References

- [HP03] W. Cary Huffman and Vera Pless. *Fundamentals of Error Correcting Codes*. Cambridge University Press, 2003.
- [KCP16] Santhosh Kumar, Robert Calderbank, and Henry D. Pfister. Beyond Double Transitivity: Capacity-Achieving Cyclic Codes on Erasure Channels. In *2016 IEEE Information Theory Workshop (ITW)*, pages 241–245, 2016.
- [KKM⁺17] Shrinivas Kudekar, Santhosh Kumar, Marco Mondelli, Henry D. Pfister, Eren Şaşoğlu, and Rüdiger L. Urbanke. Reed–Muller Codes Achieve Capacity on Erasure Channels. *IEEE Transactions on Information Theory*, 63(7):4298–4316, 2017.
- [MS77] Florence MacWilliams and Neil Sloane. *The Theory of Error Correcting Codes*. North Holland, 1977.
- [NK23] Lakshmi Prasad Natarajan and Prasad Krishnan. Berman Codes: A Generalization of Reed–Muller Codes That Achieve BEC Capacity. *IEEE Transactions on Information Theory*, 69(11):6956–6980, 2023.
- [RSU01] T.J. Richardson, M.A. Shokrollahi, and R.L. Urbanke. Design of Capacity-Approaching Irregular Low-Density Parity-Check Codes. *IEEE Transactions on Information Theory*, 47(2):619–637, 2001.

- [Sha48] C. E. Shannon. A Mathematical Theory of Communication. *The Bell System Technical Journal*, 27(3):379–423, 1948.
- [Wei24] Eric W. Weisstein. “Zsigmondy Theorem”. MathWorld – A Wolfram Web Resource, 2024.
- [Wel25] Daniel Welchons. *Using Permutation Groups to Identify Families of Capacity Achieving Codes*. PhD thesis, University of Nebraska–Lincoln, 2025.
- [Zsi82] K Zsigmondy. Zur Theorie der Potenzreste. *Monatsh. f. Mathematik und Physik*, pages 267–284, 1982.